FA-2 SLIP TEST

Class: 8 | Subject: Mathematics
Time: 45 Minutes | Total Marks: 20 Marks

Name of the Student: .....
Roll Number: .....
Date: .....

Problem 1 [4 Marks]

Evaluate using laws of exponents:

$$(2^{-1} \times 4^{-1}) \div 2^{-2}$$

Problem 2 [4 Marks]

Find the value of x such that:

$$(2/3)^{-5} \times (2/3)^{-7} = (2/3)^{2x}$$

Problem 3 [4 Marks]

Convert the ratio **3 : 4** into a percentage, and express **35%** as a ratio in simplest form.

Problem 4 [4 Marks]

A shopkeeper bought an article for **₹ 750** and spent **₹ 50** on transportation. If he sold it for **₹ 960**, calculate his profit percentage.

Problem 5 [4 Marks]

An article with marked price **₹ 1,200** is sold for **₹ 1,020**. Determine the discount given and the discount percentage.

FA-2 SLIP TEST — SOLUTIONS

Step-by-Step Marking Scheme & Detailed Working (20 Marks)

Solution 1: [4 Marks]

Step 1: Using the rule $a^{-n} = {}^{1}/_{a^n}$, convert terms into fractions.

$2^{-1} = {}^{1}/_{2}$ and $4^{-1} = {}^{1}/_{4}$

Step 2: Multiply inside the brackets: $({}^{1}/_{2} \times {}^{1}/_{4}) = {}^{1}/_{8}$

Step 3: Convert divisor: $2^{-2} = {}^{1}/_{2^2} = {}^{1}/_{4}$

Step 4: Divide: ${}^{1}/_{8} \div {}^{1}/_{4} = {}^{1}/_{8} \times {}^{4}/_{1} = {}^{4}/_{8} = {}^{1}/_{2}$

Answer = ${}^{1}/_{2}$

Solution 2: [4 Marks]

Step 1: Apply product law of exponents $a^m \times a^n = a^{m+n}$ on LHS:

$({}^{2}/_{3})^{(-5)+(-7)} = ({}^{2}/_{3})^{2x}$

Step 2: Simplify exponent on LHS: $({}^{2}/_{3})^{-12} = ({}^{2}/_{3})^{2x}$

Step 3: Since the bases are identical $({}^{2}/_{3})$, equate the exponents:

$-12 = 2x \Rightarrow x = {}^{-12}/_{2}$

Answer: x = −6

Solution 3: [4 Marks]

Step 1: Convert ratio 3 : 4 into percentage:

$({}^{3}/_{4}) \times 100\% = 3 \times 25\% = 75\%$

Step 2: Convert 35% into fraction form:

$35\% = {}^{35}/_{100}$

Step 3: Simplify by dividing numerator and denominator by 5: ${}^{35 \div 5}/_{100 \div 5} = {}^{7}/_{20}$

Answer: 75% and 7 : 20

Solution 4: [4 Marks]

Step 1: Total Cost Price (CP) = Purchase Price + Overhead Charges

CP = ₹ 750 + ₹ 50 = ₹ 800

Step 2: Given Selling Price (SP) = ₹ 960

Step 3: Profit = SP − CP = ₹ 960 − ₹ 800 = ₹ 160

Step 4: Profit % = (Profit / CP) × 100 = $({}^{160}/_{800}) \times 100 = 20\%$

Answer: Profit = 20%

Solution 5: [4 Marks]

Step 1: Marked Price (MP) = ₹ 1,200 | Selling Price (SP) = ₹ 1,020

Step 2: Discount = MP − SP = ₹ 1,200 − ₹ 1,020 = ₹ 180

Step 3: Discount % = (Discount / MP) × 100

Discount % = $({}^{180}/_{1200}) \times 100 = {}^{180}/_{12} = 15\%$